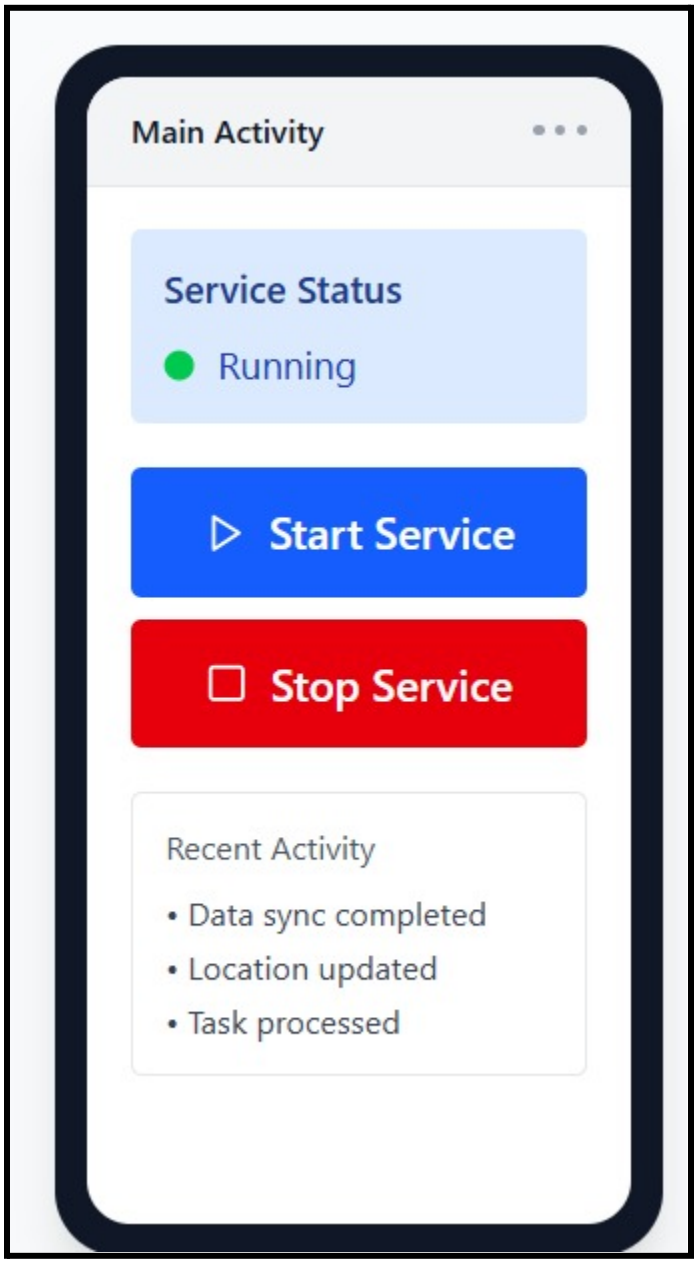
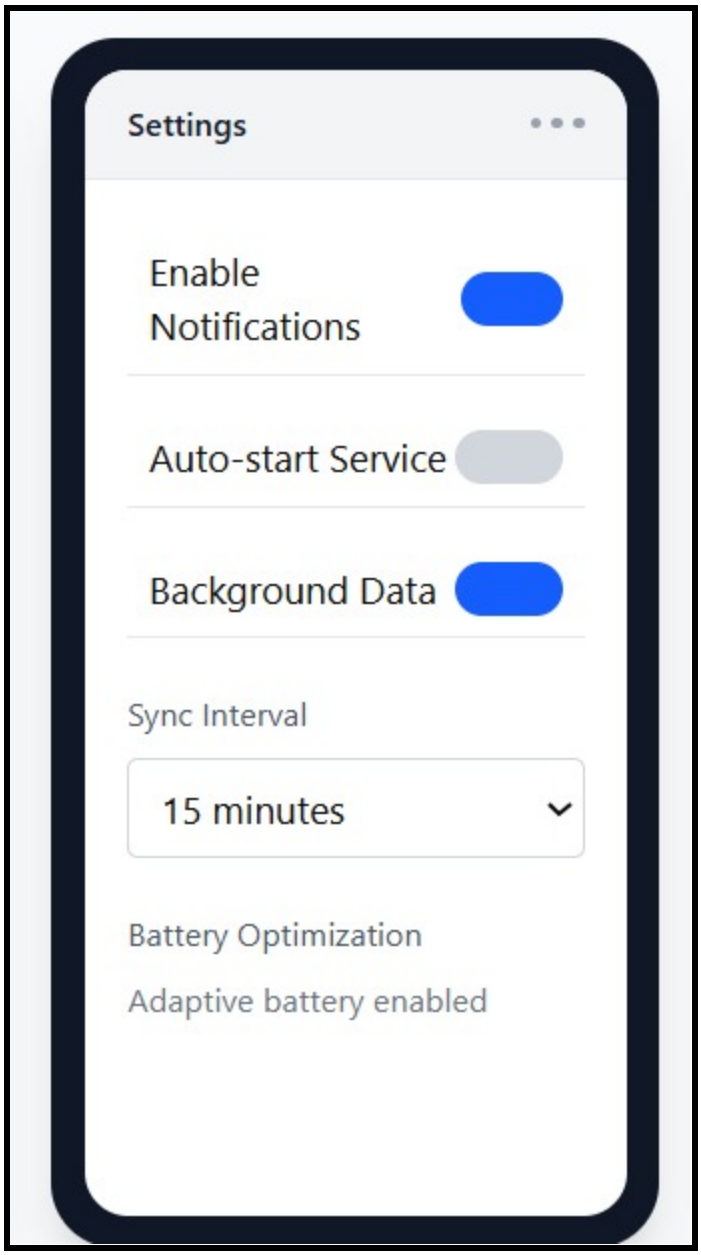
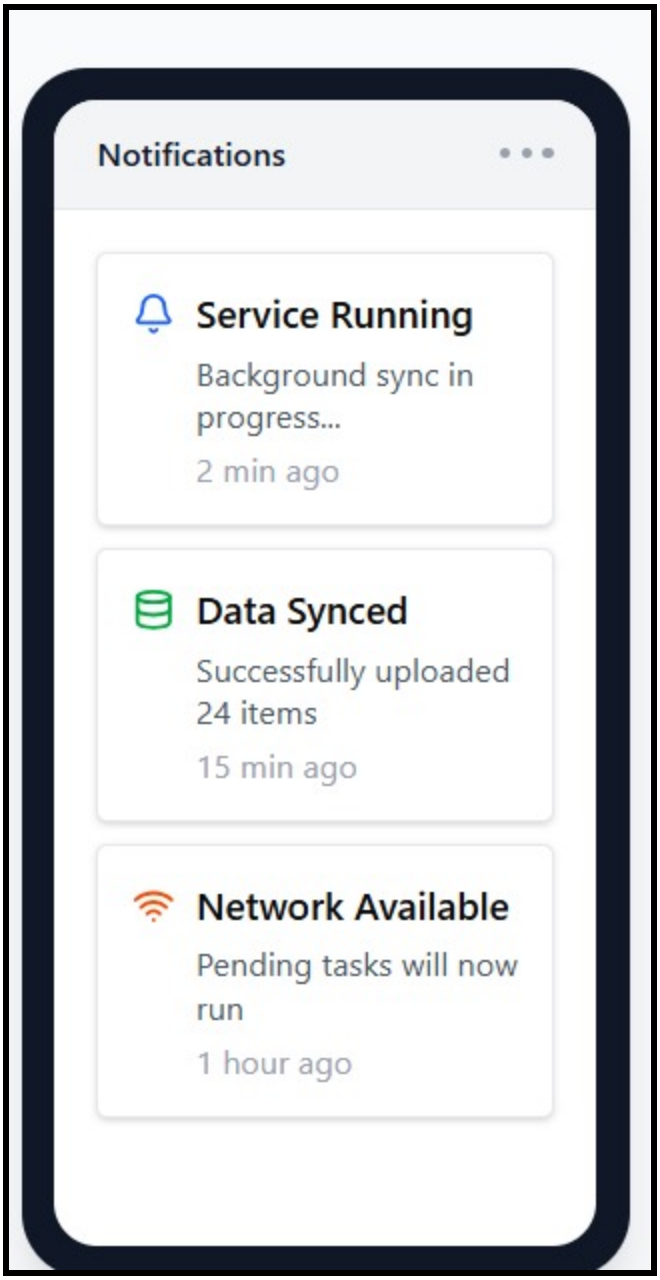
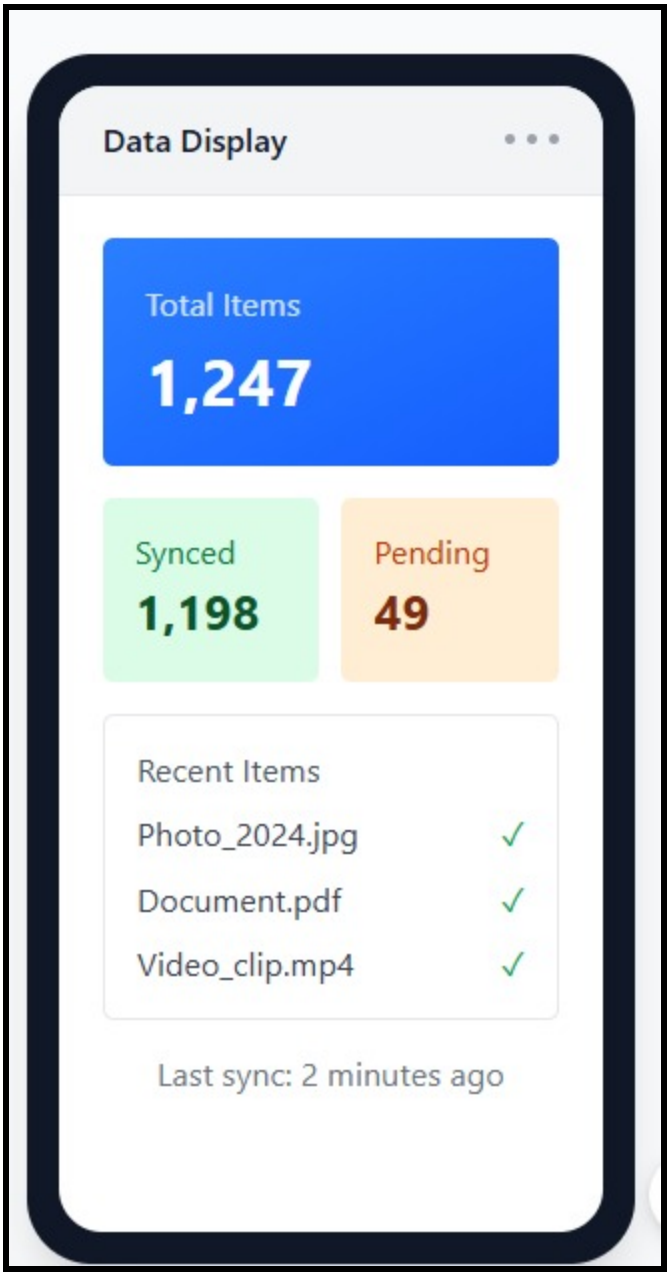
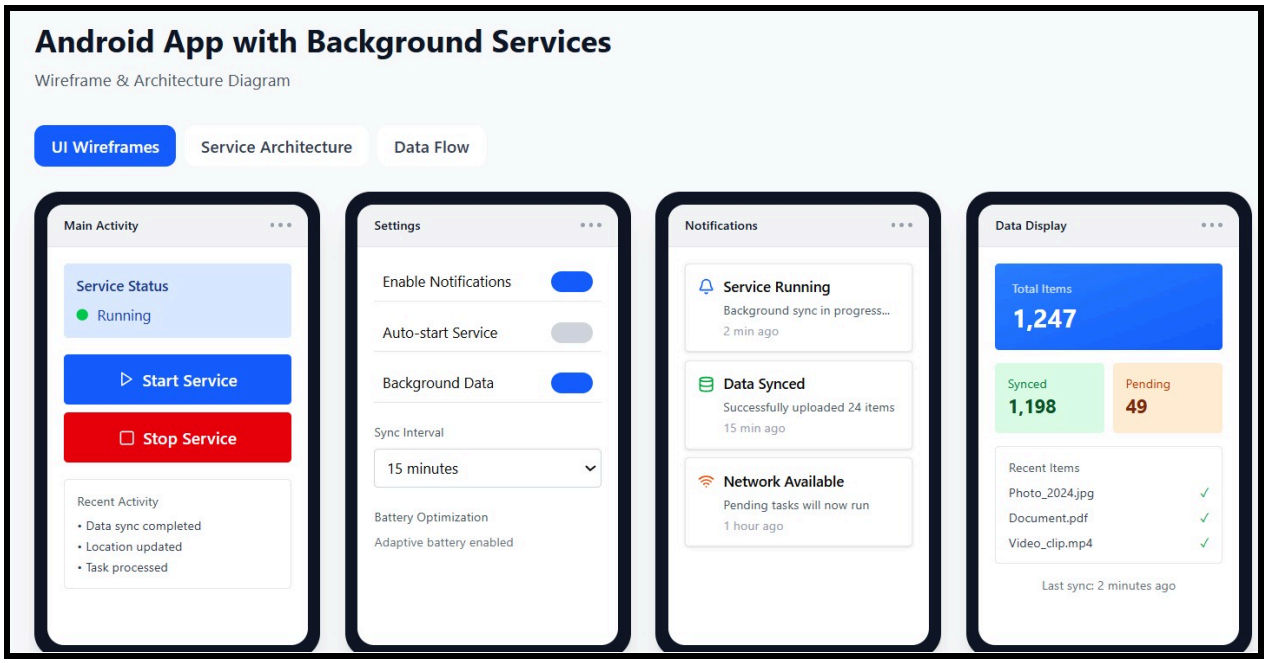

Main Activity
Service Status
Running
Start Service
Stop Service
Recent Activity
• Data sync completed
• Location updated
• Task processed


Settings
Enable Notifications
Auto-start Service
Background Data
Sync Interval
15 minutes
Battery Optimization
Adaptive battery enabled


Notifications
Service Running
Background sync in progress...
2 min ago
Data Synced
Successfully uploaded 24 items
15 min ago
Network Available
Pending tasks will now run
1 hour ago


Data Display
Total Items
1,247
Synced
1,198
Pending
49
Recent Items
Photo_2024.jpg
Document.pdf
Video_clip.mp4
Last sync: 2 minutes ago


Android App with Background Services
Wireframe & Architecture Diagram
UI Wireframes
Service Architecture
Data Flow
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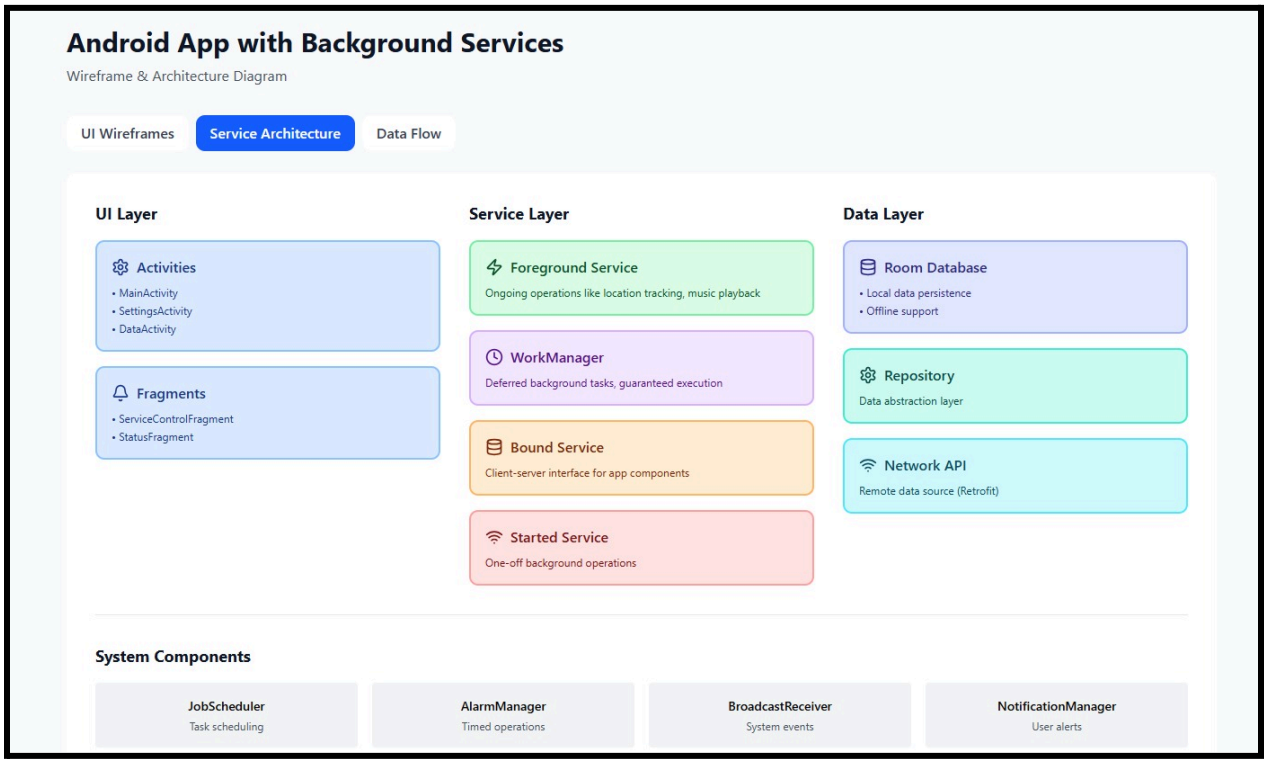
Android App with Background Services
Wireframe & Architecture Diagram
UI Wireframes
Service Architecture
Data Flow
UI Layer
Activities
• MainActivity
• SettingsActivity
• DataActivity
Fragments
• ServiceControlFragment
• StatusFragment
Service Layer
Foreground Service
Ongoing operations like location tracking, music playback
WorkManager
Deferred background tasks, guaranteed execution
Bound Service
Client-server interface for app components
Started Service
One-off background operations
Data Layer
Room Database
• Local data persistence
• Offline support
Repository
Data abstraction layer
Network API
Remote data source (Retrofit)
System Components
JobScheduler
Task scheduling
AlarmManager
Timed operations
BroadcastReceiver
System events
NotificationManager
User alerts

Android App with Background Services

Wireframe & Architecture Diagram

UI Wireframes | Service Architecture | **Data Flow**

1. Service Initialization

User Action	→	MainActivity	
MainActivity	→	startService() Intent	
Intent	→	BackgroundService.onCreate()	
BackgroundService	→	Notification	*Show foreground notification*

2. Background Data Processing

BackgroundService	→	Repository	
Repository	→	Network API	*Fetch remote data*
Network API	→	Repository	*Return data*
Repository	→	Room Database	*Cache locally*
Room Database	→	Repository	*Confirm save*